

AGHASI KHACHATRYAN

Senior Backend Engineer · Java · Go · Spring Boot · Distributed Systems · Cloud
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PROFESSIONAL SUMMARY

Senior Backend Engineer with 6+ years in Java and Go. Builds distributed systems: microservices, REST and gRPC APIs, event-driven pipelines. Commands Spring Boot, Micronaut, JVM concurrency and clean architecture in production systems. Designs scalable architecture for fintech, banking, cloud-infrastructure and e-commerce platforms. Delivers on AWS with Docker, Kubernetes, Helm and CI/CD pipelines. Models data across relational and NoSQL stores, with advanced SQL and query tuning. Owns systems from design decision through deployment to production support. Leads code review, drives core system-design decisions and mentors engineers. Currently builds OMD's tick-level market-data platform; previously led architecture decisions and code review across e-commerce teams. Uses Claude Code and OpenCode CLI in daily production development.

KEY ACHIEVEMENTS

- Led the migration of a monolithic e-commerce platform to microservices, then built white-label-capable products on the new architecture.
- Architected the reactive Java 21 / Micronaut services and the Go concurrency module behind a tick-level market-data platform.
- Delivered four production backend products as sole or lead developer, including a full B2C e-commerce platform.
- Drove team-wide adoption of Claude Code and LLM-assisted review, keeping human code review as the release gate.

CORE SKILLS

Languages Java (8 / 11 / 17 / 21), Go (Golang), SQL, Python (automation)

Frameworks Spring (Boot 3.x, Security, Cloud, Data), Hibernate, JPA, Micronaut, Micronaut Data, Apache Camel, Uber FX

Architecture Distributed systems, scalable architecture, microservices, MVC, REST APIs, gRPC / Protocol Buffers (Protobuf), event-driven & streaming systems, reactive services, concurrency / multithreading, high availability, performance optimization, SOLID, Clean Architecture, design patterns, contract-first OpenAPI

Messaging Apache Kafka, ActiveMQ, AWS SQS, WebSockets, Server-Sent Events

Data PostgreSQL, MySQL, Oracle, H2 — schema design, advanced SQL, indexing & query tuning · MongoDB, Redis, Elasticsearch · Flyway, Liquibase · R2DBC, JDBC, MyBatis

Security Spring Security, OAuth2, JWT (JSON Web Token) lifecycle, role-based access control (RBAC), Keycloak, PASETO

Cloud & DevOps AWS (EKS, S3, SQS, IAM, SSO / single sign-on, EC2) · Docker, Kubernetes, Helm · CI/CD pipelines (continuous integration and delivery): GitHub Actions, GitLab CI · Nginx, Cloudflare, Linux

Testing & Performance JUnit 5, Mockito, AssertJ, integration & end-to-end test automation · p95 / p99 latency tuning, query and throughput profiling

Observability Structured logging, Grafana dashboards and performance metrics, high availability, production debugging and incident response

AI Engineering Agentic coding tools (Claude Code, OpenCode CLI) in daily development · Anthropic API integration · LLM-assisted refactoring, test generation, code review and documentation · prompt engineering

Leadership Team leadership, technical direction and system-design ownership, code review, pull-request review, team collaboration, mentoring and training engineers

Practices Git & branching workflows, Agile / Scrum, Maven, technical documentation

PROFESSIONAL EXPERIENCE

OMD — Tick Data

Sep 2024 – Present

Senior Java Software Engineer

Financial data analytics platform delivering historical market data, tick-level processing and analytics for trading and quantitative research.

- Architected reactive microservices in Java 21 and Micronaut with GraalVM native builds, powering tick-level market-data processing over Micronaut Data, R2DBC and JDBC.
- Designed REST APIs for internal and external consumers, including pageable, searchable endpoints with server-side filtering and sorting.
- Engineered streaming file downloads that deliver large files as they are generated, reworking the generation pipeline to cut wait times and memory footprint.
- Integrated asynchronous messaging over ActiveMQ and AWS SQS, plus real-time WebSocket channels, for decoupled event-driven services.
- Secured the API surface: authentication, authorization, JWT token lifecycle and role-based access control.
- Built a custom API gateway, a centralized configuration service with Caffeine caching and dynamic reload, and data-migration services for scheduled and on-demand runs.
- Own a Go platform module built on goroutines, worker-pool patterns and sync primitives for high-throughput concurrent data processing.
- Standardized platform API contracts with contract-first OpenAPI and code generation, and authored reusable validation and exception-handling libraries now used by every service team.
- Led team-wide adoption of Claude Code and OpenCode CLI to accelerate refactoring, test generation and documentation, keeping human review as the final quality gate.
- Instrumented services with structured logging and Grafana metrics, giving the team observability into production behaviour.
- Drove scalability and performance optimization through targeted refactoring and query tuning, tracking p95 / p99 latency to locate slow paths.
- Backed every change with unit, integration and end-to-end test coverage.
- Drive core system-design decisions for the platform's Java services, mentor engineers through code review and own support for legacy versions.

Technologies: Java 21, Go, GraalVM, Micronaut / Micronaut Data, R2DBC, JDBC, PostgreSQL, MySQL, H2, Flyway, ActiveMQ, WebSockets, Caffeine, OpenAPI, REST, Docker, Kubernetes, Helm, AWS (EKS, SQS, S3, IAM, SSO), Python, GitLab CI/CD, JUnit 5, AssertJ, Mockito, Grafana, Git, Linux

Energize Global Services Senior Java Software Engineer <i>PGUI — bank management system.</i>		Jul 2023 – Jul 2024
- Engineered new features and resolved production incidents on a bank-management platform, sustaining service reliability. - Designed REST APIs and built user-management and search functionality, backed by Keycloak for identity and access management. - Tuned slow SQL queries to improve database performance and response times, and maintained schema integrity through versioned Liquibase migrations. Technologies: Java 11, Spring Boot, Tapestry, Keycloak, Apache Camel, Oracle, Hibernate, Liquibase, Maven, REST, Git		
I-Tech Java Backend Developer <i>Audit Trail (user-activity logging), T3 TMS (transportation management), TMT (GPS trip tracking), Click Market (B2C e-commerce).</i>		Aug 2022 – Jul 2023
- Delivered four production backend products as sole or lead developer on Spring Boot 3 and Spring Cloud. - Built REST APIs, JWT security, search and notification features across those services. - Built an audit-logging service on Elasticsearch and MongoDB that consumed Kafka events, tracking p95 / p99 query latency as the index grew. - Shipped Click Market, a B2C e-commerce platform, as sole developer — payment integration, catalog, cart and checkout, admin dashboard and inventory logic. - Engineered S3-backed file uploads and SSE streaming, and owned database migrations. - Architected the multi-module application and database structure, then built the CI/CD pipelines and containerized every service with Docker. - Owned code review and automated test coverage across the delivered services. Technologies: Java 17/11, Spring Boot 3, Spring Security, Spring Cloud, Spring Data (JPA / Elasticsearch / MongoDB), Kafka, Elasticsearch, MongoDB, MySQL, PostgreSQL, SSE, JUnit 5, Docker, Kubernetes, CI/CD, Hibernate, Liquibase, AWS S3, Git		
Codeex Java Software Engineer <i>MLM — multi-level marketing e-commerce platform.</i>		Aug 2020 – Aug 2022
- Migrated a monolithic application to microservices and built white-label-capable products. - Developed a PASETO-secured authorization server and the e-commerce microservices behind it, and designed the user-management database architecture. - Led code reviews and architecture decisions across the team. Technologies: Java 17, Spring Boot, Spring Security, Spring Cloud, PostgreSQL, Redis, WebSockets, Hibernate, MyBatis, Liquibase, Docker, Maven, Git		
Smart Code (Part-time) Java Trainer (Part-time) — concurrent with the Energize and OMD roles		Jul 2023 – Sep 2024
- Trained aspiring Java developers in core Java, Spring and backend engineering practice. - Authored the course materials and hands-on exercises covering OOP, data structures, algorithms and clean code.		
PicsArt Armenia Java Developer Trainee		Feb 2020 – Aug 2020
Completed an intensive Java backend training program and contributed to production development tasks under senior mentorship.		
SELECTED ADDITIONAL PROJECT		
SoulsHub (Freelance) Architecture & DevOps Engineer <i>SoulsHub — an LLM-powered platform for interacting with AI representations of famous personalities, historical figures and mentors.</i>		Jan 2024
- Designed resilient application and database architecture for an LLM-based conversational product. - Configured Nginx and Cloudflare for routing, TLS termination and edge protection. - Built GitHub Actions CI/CD: automated builds, quality gates, approval steps and Docker image deploys to staging and production. - Provisioned and hardened AWS EC2 instances with security groups and IAM roles. - Automated S3 database backups with scheduled jobs and version-based rollback. Technologies: GitHub Actions, GitHub Container Registry, Nginx, Cloudflare, Docker, AWS (EC2, S3, IAM), Linux, Git		
CERTIFICATIONS & TRAINING		
AWS Cloud Practitioner Essentials — Amazon Web Services, Feb 2023 · Credential ID 1CD4437E-4111-40EB-A4E1-54105F6F3C6F AWS Cloud Essentials — Amazon Web Services, Feb 2023 · Credential ID 189192CE-ADCC-406C-BC2A-8EEBB954F784 Java Development — Basic IT Center, Sep 2020 Java Backend Trainee Program — PicsArt Armenia, 2020; intensive full-time backend program. Java & Spring Backend Curriculum — Smart Code, 2023–2024; authored and taught as instructor.		
EDUCATION & LANGUAGES		
BSc, Informatics and Computer Science National University of Architecture and Construction of Armenia (NUACA)		2014 – 2018
Languages English (B2 upper-intermediate — professional working proficiency) · Russian (fluent) · Armenian (native)		